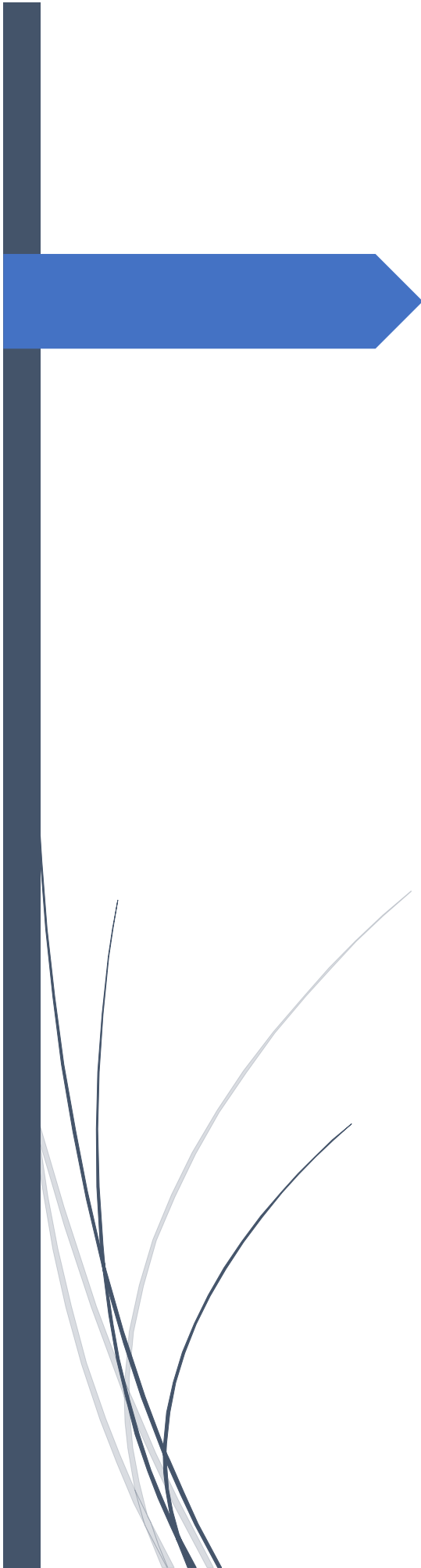




Optional Assignment

Configure ASA Basic Settings and Firewall Using the CLI

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CS-CNS09-25030

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Introduction

In modern enterprise networks, firewalls serve as the first line of defense against unauthorized access and potential threats. This practical exercise focused on configuring a Cisco Adaptive Security Appliance (ASA) using the command-line interface (CLI) within Cisco Packet Tracer. The scenario simulated a small business network with internal, external, and DMZ segments, where the ASA was configured to provide routing, NAT, DHCP, SSH access, and firewall protection. By methodically applying each configuration—ranging from interface setup to implementing AAA authentication and DMZ access control—the goal was to build a secure, functional network perimeter that supports both internal connectivity and controlled external access. This exercise not only reinforced ASA command syntax but also deepened my understanding of real-world firewall deployment strategies, zone-based security, and access control mechanisms.

Part 1

Verifying Connectivity and Explore the AS

Verifying connectivity

To begin the configuration process, I first verified basic connectivity across the network before any ASA configurations were made. I started by logging into PC-C and successfully pinged the router interfaces, such as 10.2.2.1 on R3 and 10.1.1.1 on R1, confirming that the upstream network was reachable. However, as expected, I was unable to ping the ASA interface (both 192.168.1.1 and 209.165.200.226), PC-B, or the DMZ server since the ASA was not yet configured to allow such traffic. This validated that the ASA was acting as an unconfigured firewall and no NAT or routing was in place.

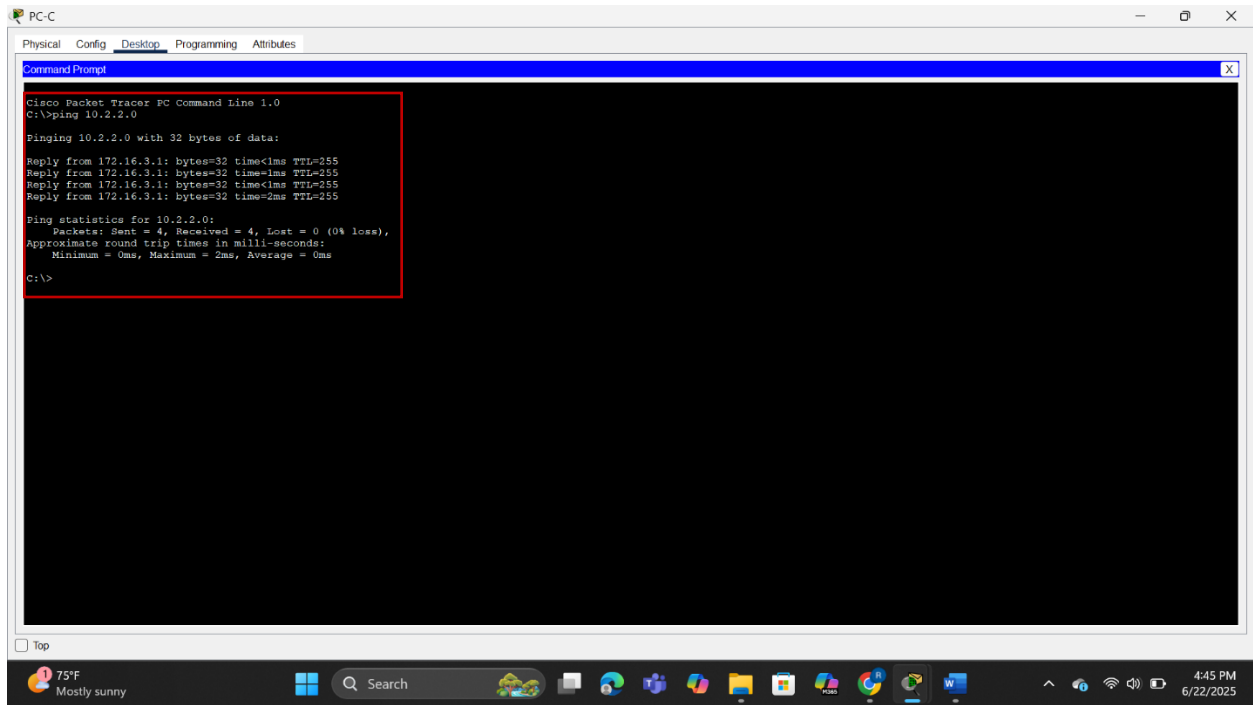


Fig 1.0 Pinging R3 from pc C

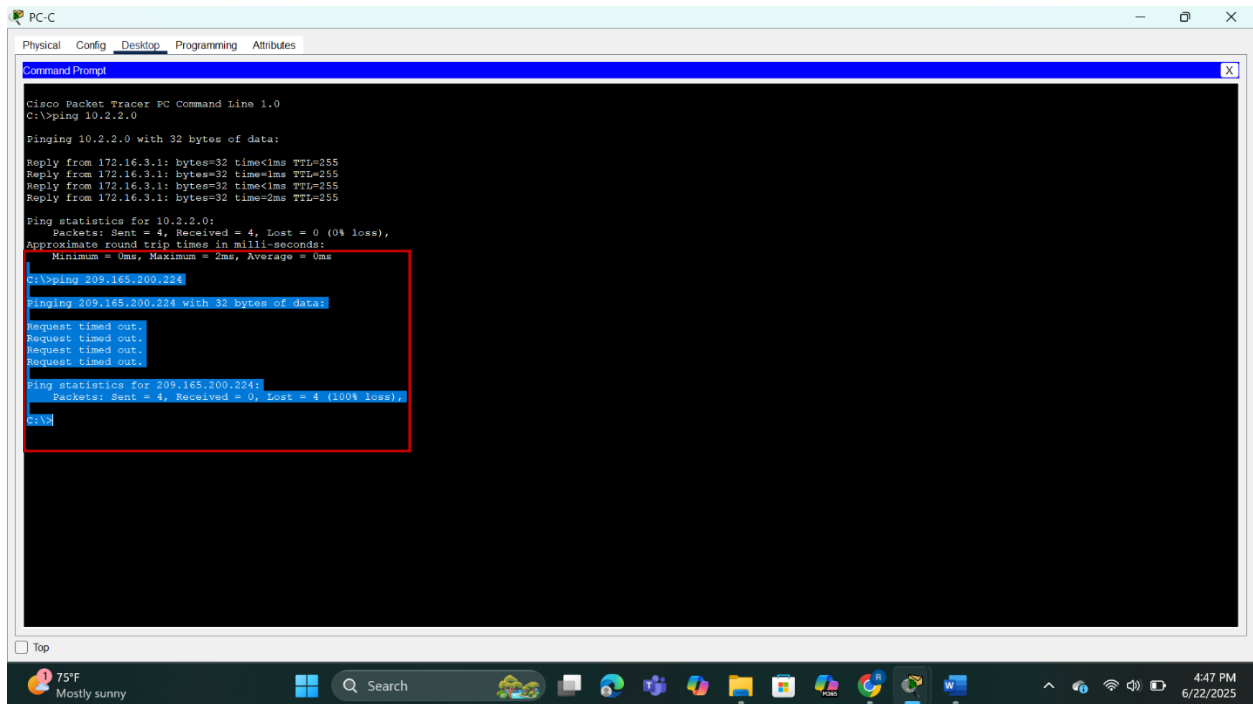


Fig 1.1 pinging ASA from pc C

```
PC-C
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.2.2.0

Pinging 10.2.2.0 with 32 bytes of data:

Reply from 172.16.3.1: bytes=32 time<1ms TTL=255
Reply from 172.16.3.1: bytes=32 time<1ms TTL=255
Reply from 172.16.3.1: bytes=32 time<1ms TTL=255
Reply from 172.16.3.1: bytes=32 time<2ms TTL=255

Ping statistics for 10.2.2.0:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 209.165.200.224

Pinging 209.165.200.224 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 209.165.200.224:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.1.1.0

Pinging 10.1.1.0 with 32 bytes of data:

Reply from 10.2.2.2: bytes=32 time=30ms TTL=254
Reply from 10.2.2.2: bytes=32 time=2ms TTL=254
Reply from 10.2.2.2: bytes=32 time=1ms TTL=254
Reply from 10.2.2.2: bytes=32 time=1ms TTL=254

Ping statistics for 10.1.1.0:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 30ms, Average = 8ms

C:\>
```

Fig 1.2 pinging R1 from pc C

```
PC-C
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 172.16.3.1: bytes=32 time=2ms TTL=255

Ping statistics for 10.2.2.0:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 209.165.200.224

Pinging 209.165.200.224 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 209.165.200.224:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 10.1.1.0

Pinging 10.1.1.0 with 32 bytes of data:

Reply from 10.2.2.2: bytes=32 time=30ms TTL=254
Reply from 10.2.2.2: bytes=32 time=2ms TTL=254
Reply from 10.2.2.2: bytes=32 time=1ms TTL=254
Reply from 10.2.2.2: bytes=32 time=1ms TTL=254

Ping statistics for 10.1.1.0:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 30ms, Average = 8ms

C:\>ping 192.168.1.0

Pinging 192.168.1.0 with 32 bytes of data:

Reply from 10.2.2.2: Destination host unreachable.
Reply from 10.2.2.2: Destination host unreachable.
Reply from 10.2.2.2: Destination host unreachable.
Reply from 10.2.2.2: Destination host unreachable.

Ping statistics for 192.168.1.0:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Fig 1.3 Pinging pc B from pc C

Determining the ASA version, interfaces, and license

Next, I accessed the ASA device via the CLI and entered privileged EXEC mode by typing enable, which did not require a password. I then used the show version command to examine the ASA's software version, license type, and active interfaces. This confirmed the ASA was running and had its physical interfaces recognized, though not yet configured.

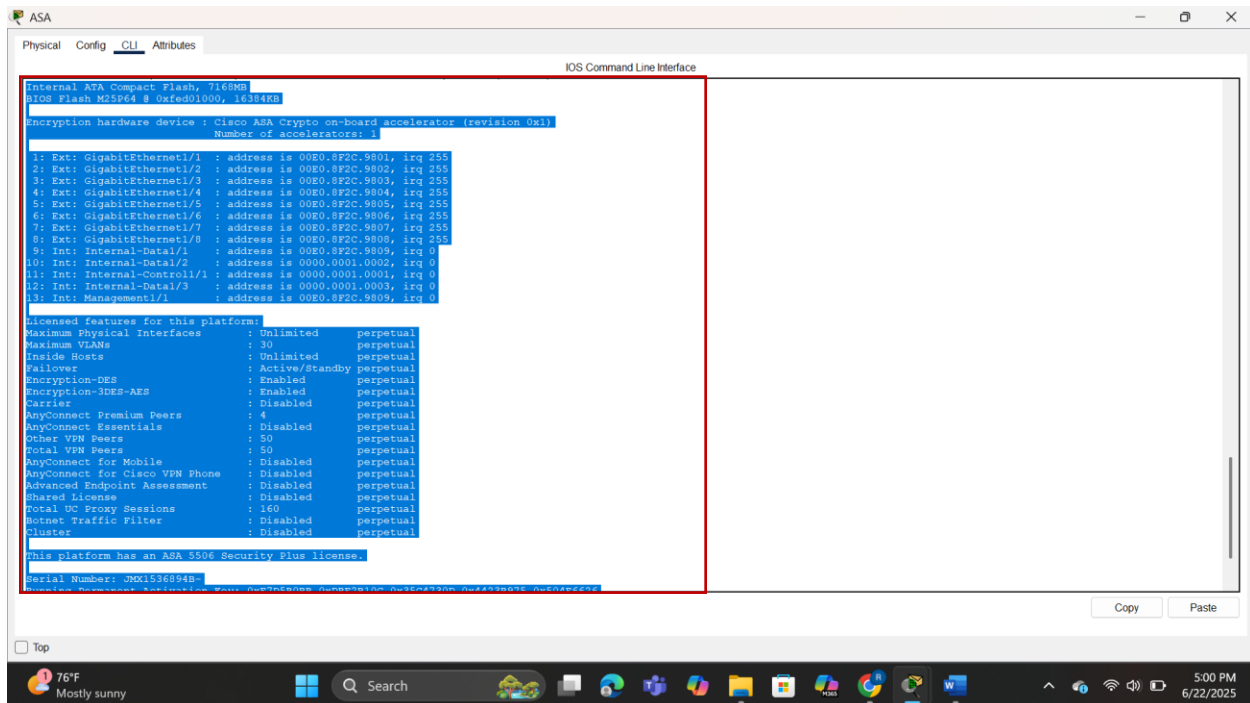


Fig 1.4 ASA's software version

Determining the file system and contents of flash memory

Finally, I used show file system to examine the file systems supported by the ASA, identifying common prefixes like disk0: for flash memory. I followed this by using show flash: to inspect the files present in ASA's flash memory, confirming the presence of core system files needed for booting and operation. This completed my preliminary system checks and provided a foundational understanding of the ASA's current status prior to configuration.

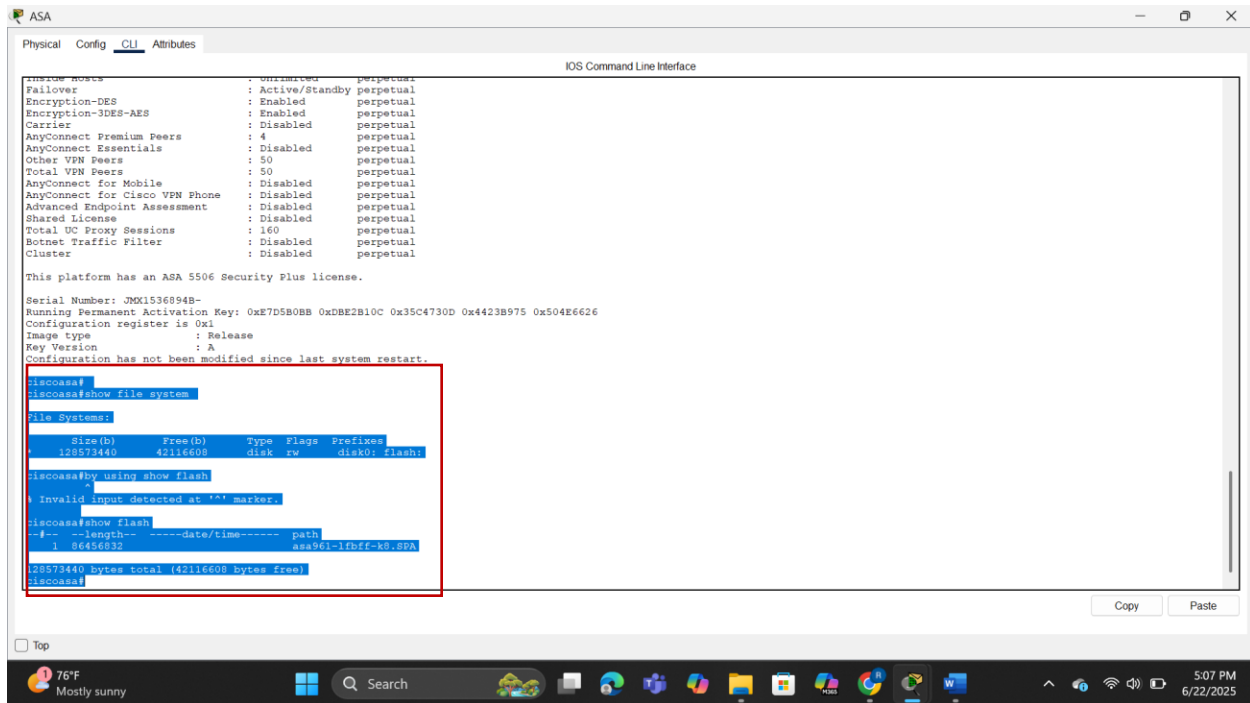


Fig 1.5 file systems supported by the ASA

Part 2

Configuring ASA Settings and Interface Security Using the CLI

Configuring hostname and domain name

I entered global configuration mode on the ASA and set a unique hostname using the command `hostname NETSEC-ASA`, which helps in identifying the device on the network. Then, I configured the domain name to `netsec.com` using `domain-name netsec.com`. This domain name is essential later for generating RSA keys for SSH and other cryptographic functions that require a fully qualified domain name (FQDN).

The screenshot shows the ASA Command Line Interface with the following commands and output:

```
Anyconnect for Cisco VPN Phone : Disabled perpetual
Advanced Endpoint Assessment : Disabled perpetual
Shared License : Disabled perpetual
Total UC Proxy Sessions : 160 perpetual
Botnet Traffic Filter : Disabled perpetual
Cluster : Disabled perpetual

This platform has an ASA 5506 Security Plus license.
Serial Number: JMX15368948-
Running Permanent Activation Key: 0x27D5B0BB 0xDBE2B10C 0x35C4730D 0x4423B975 0x504E6626
Configuration register is 0x1
Image type : Release
Key Version : A
Configuration has not been modified since last system restart.

ciscoasa#
ciscoasa#show file system

File Systems:
      Size(b)      Free(b)      Type  Flags  Prefixes
* 128573440      42116608      disk  rw    disk0: flash:

ciscoasa#by using show flash
^
% Invalid input detected at '^' marker.

ciscoasa#show flash
--#-- --length-- -----date/time----- path
* 128573440 42116608 06/22/2025 18:04:40 k9.SPA

128573440 bytes total (42116608 bytes free)
ciscoasa#
ciscoasa(config)#hostname
ciscoasa(config)#hostname
^ Incomplete command
ciscoasa(config)#hostname NETSEC-ASA
NETSEC-ASA(config)#domain name netsec.com
ciscoasa(config)#
^ Invalid input detected at '^' marker.
NETSEC-ASA(config)#domain-name netsec.com
NETSEC-ASA(config)#
```

The bottom part of the terminal window shows the configuration steps for setting the hostname and domain name, which are highlighted with a red box in the original image. The steps are:

- `ciscoasa(config)#hostname`
- `ciscoasa(config)#hostname` (incomplete command)
- `ciscoasa(config)#hostname NETSEC-ASA`
- `NETSEC-ASA(config)#domain name netsec.com`
- `NETSEC-ASA(config)#`
- `NETSEC-ASA(config)#domain-name netsec.com`
- `NETSEC-ASA(config)#`

Fig 1.6 configuring domain name and hostname

Configuring enable mode password

To secure privileged EXEC access on the ASA, I used the enable password ciscoenpa55 command. This sets the password required to enter enable mode, which allows configuration and diagnostic commands.

Configuring the INSIDE and OUTSIDE interfaces

In this step, I began by configuring the G1/1 interface on the ASA, which serves as the connection to the outside network. I entered interface configuration mode using the interface g1/1 command and assigned it the name "OUTSIDE" with the nameif OUTSIDE command. I then configured its IP address as 209.165.200.226 255.255.255.248 and set its security level to 0 using the security-level 0 command, indicating it is an untrusted external interface. Finally, I enabled the interface using the no shutdown command. Following that, I configured the G1/2 interface for the internal LAN. I accessed its interface mode using interface g1/2, named it "INSIDE" via nameif INSIDE, and assigned it the IP address 192.168.1.1 255.255.255.0. I then set its security level to 100 with security-level 100, which represents a highly trusted internal interface, and enabled it using no shutdown. These assigned security levels define the ASA's behavior in handling traffic between interfaces: by default, traffic from a higher security level (like INSIDE) to a lower one (like OUTSIDE) is permitted, whereas traffic from a lower level to a higher one is denied unless specifically allowed through configuration.

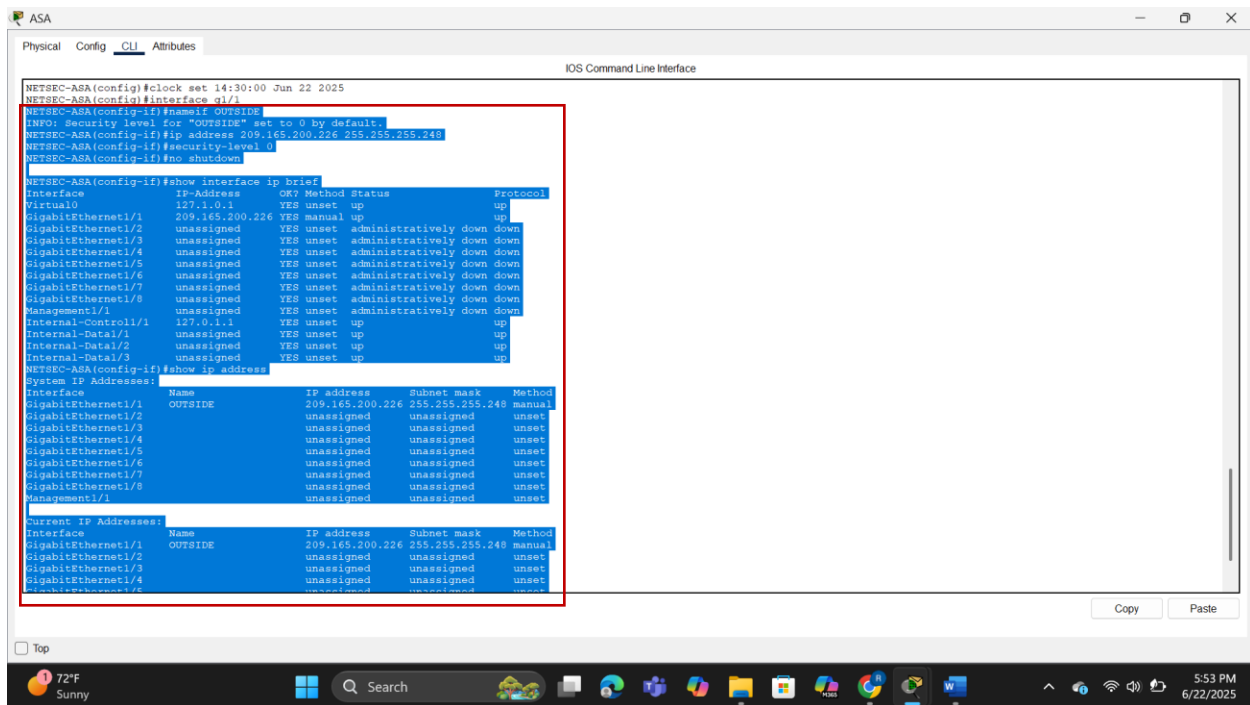


Fig 1.9 configuring outside interface

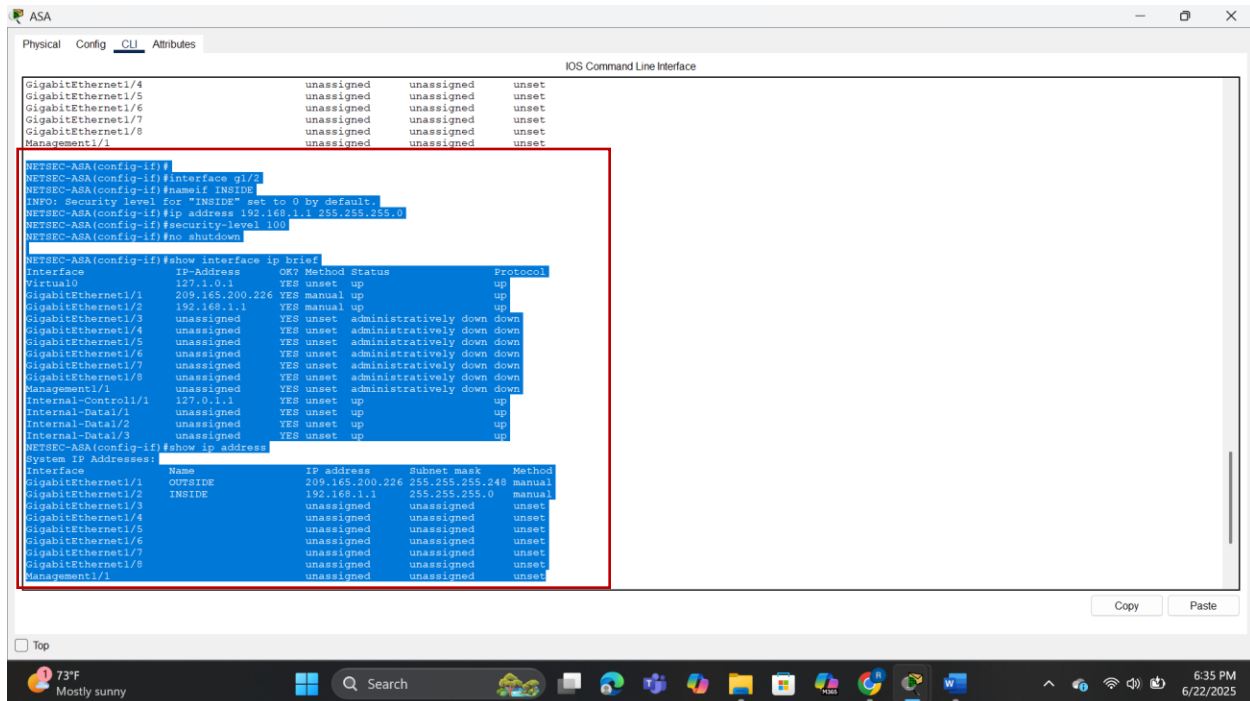


Fig 2.0 configuring inside interface and verifying

Testing connectivity to the ASA

From **PC-B**, which is connected to the INSIDE interface, I pinged **192.168.1.1** (the ASA's internal IP). The pings succeeded, confirming that the interface was configured and working. Then, I attempted to ping **209.165.200.226** (the ASA's OUTSIDE IP) from the same PC. This failed, as expected, because ASA by default does not respond to ping requests on interfaces with lower security levels unless explicitly allowed, reinforcing its role as a secure firewall.

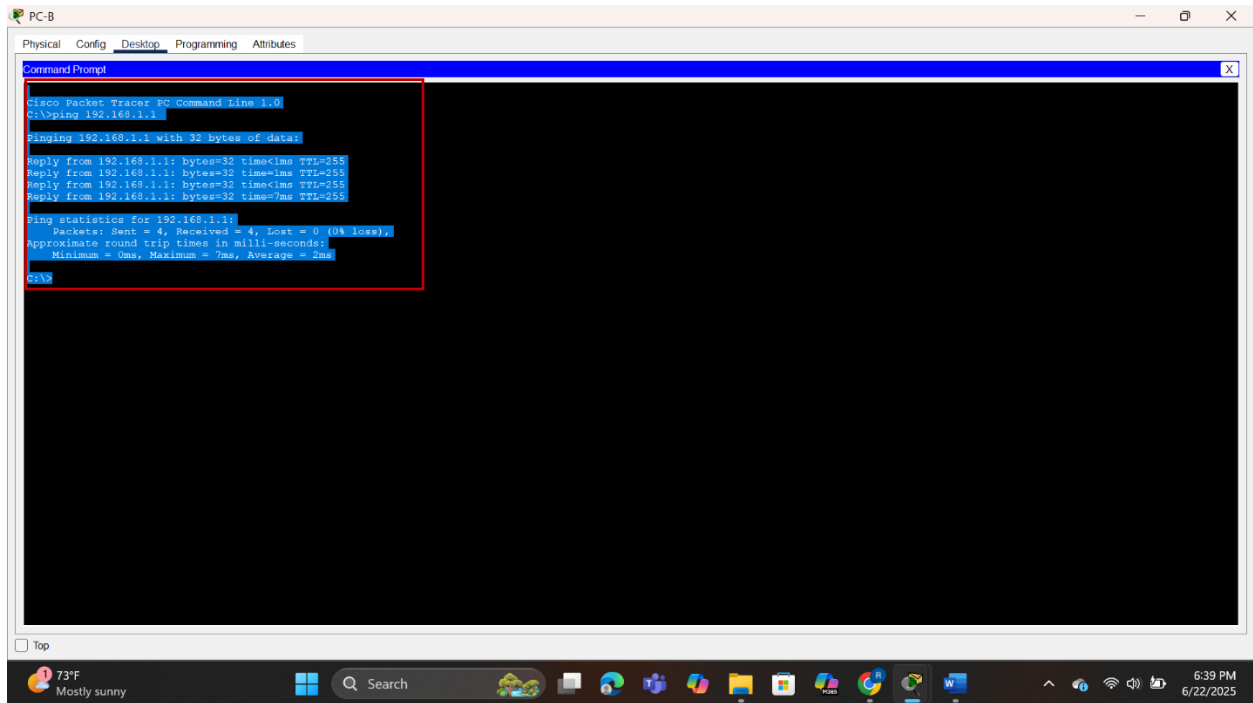


Fig 2.1 pinging inside interface

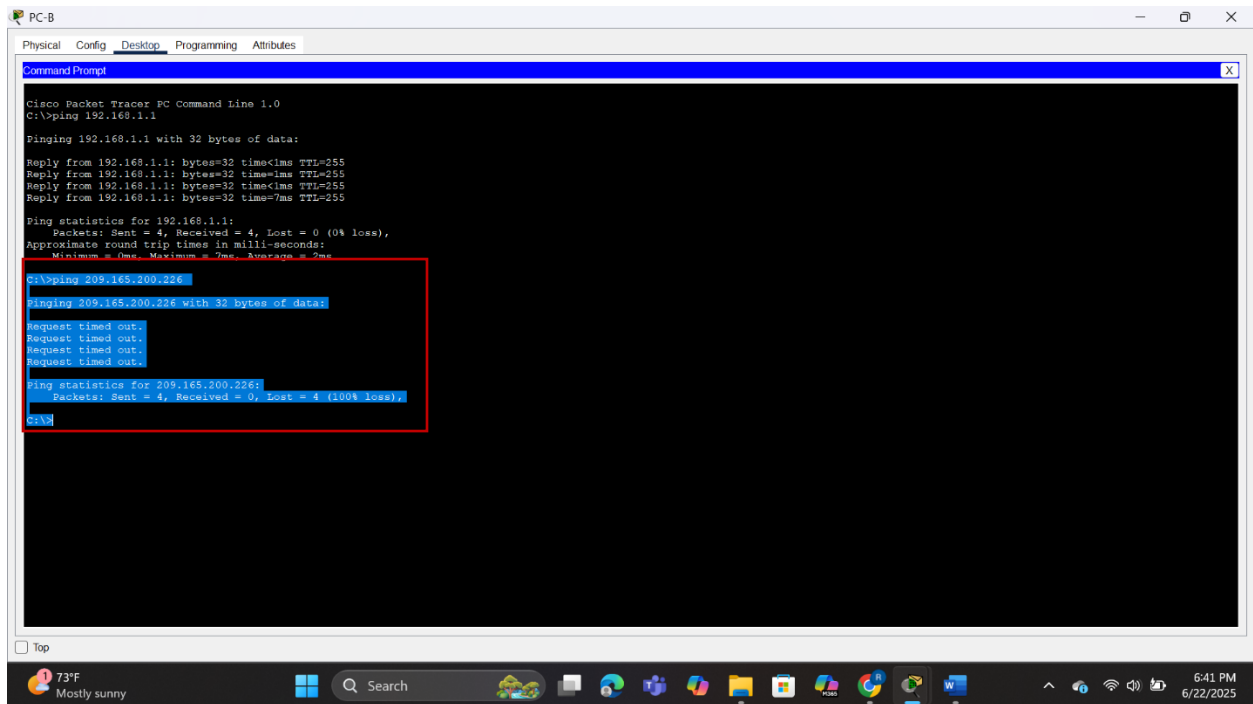


Fig 2.2 pinging outside interface

Part 3

Configuring Routing, Address Translation, and Inspection Policy Using the CLI

Configuring static default route for the ASA

To begin configuring routing on the ASA, I established a default route that directs all traffic destined for unknown networks through the ASA's OUTSIDE interface. In global configuration mode, I used the command `route OUTSIDE 0.0.0.0 0.0.0.0 209.165.200.225`, which tells the ASA to send any traffic not matching a more specific route to the next-hop IP address 209.165.200.225, which is the IP address of the R1 router's G0/0 interface. After setting this up, I verified the route's presence in the ASA's routing table using the `show route` command. The route appeared as expected, confirming that the ASA would now forward packets to external networks through its designated gateway. To further validate connectivity, I issued a ping to R1's Serial 0/0/0 interface using the IP address 10.1.1.1, and the successful replies confirmed the ASA could reach upstream networks.

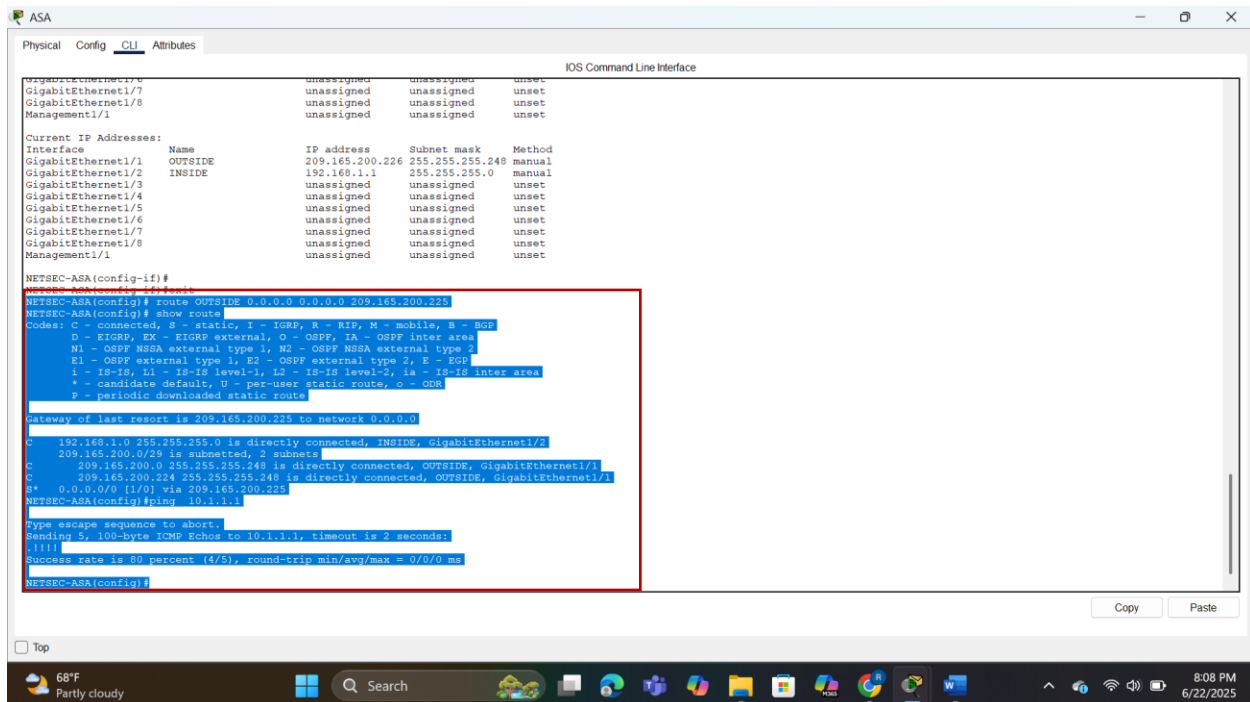
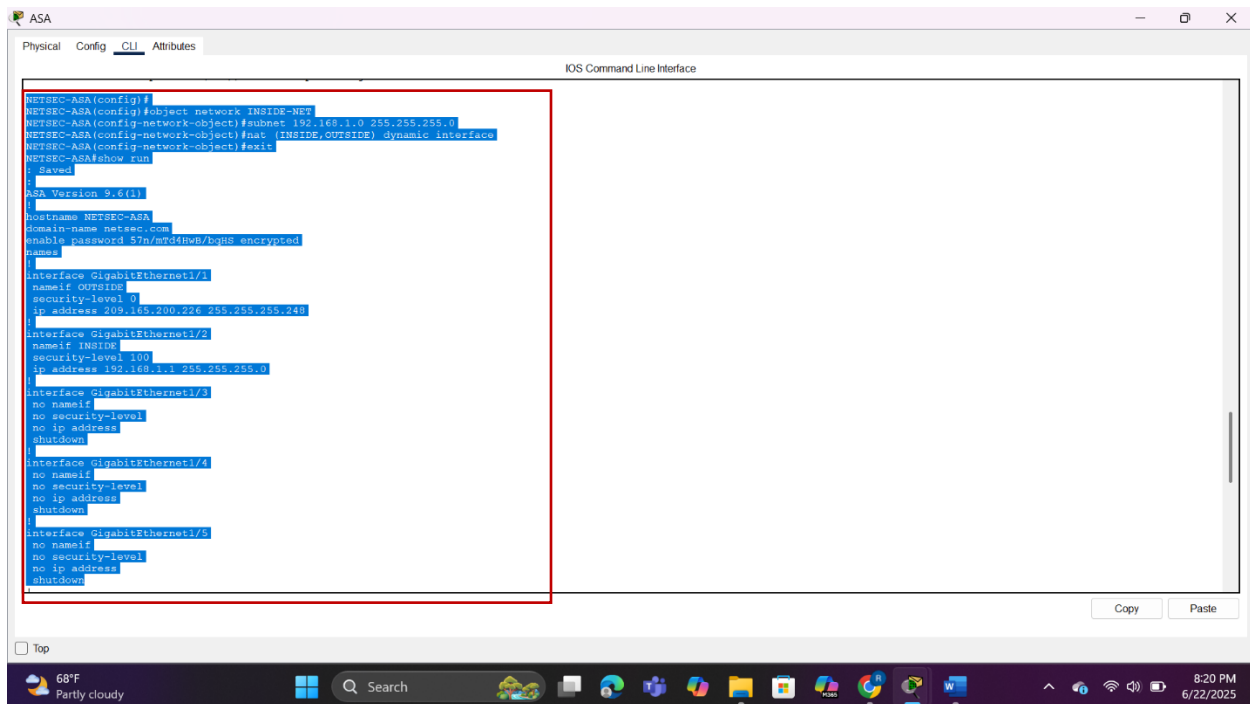


Fig 2.3 establishing a default route, verifying routes presence and pinging R1

Configuring address translation using PAT and network objects

Next, I configured address translation using PAT (Port Address Translation) with network objects. I created a network object named INSIDE-NET by entering object network INSIDE-NET, and while in object configuration mode, I defined the internal subnet using subnet

192.168.1.0 255.255.255.0. To configure PAT, I added the NAT rule nat (INSIDE,OUTSIDE) dynamic interface to allow internal hosts to be translated to the ASA's external IP when accessing the internet. This setup enables all devices in the 192.168.1.0/24 network to share the ASA's external interface IP 209.165.200.226 when making outbound connections. I confirmed the NAT configuration was successfully applied by using the show run command, which displayed both the network object definition and the corresponding NAT rule. To test whether the NAT was functioning, I initiated a ping from PC-B, which is on the INSIDE network, to R1's external IP at 209.165.200.225. As anticipated, the ping did not receive a reply due to ASA's security policies blocking return traffic. However, I used the show nat command to verify that translation hits had occurred, and indeed I observed a number of translated hits, indicating that PC-B's ICMP requests were being translated as intended—even though replies were dropped due to the default inspection policy. This confirmed the PAT configuration was successfully processing outbound traffic while enforcing ASA's default restrictive security policies.



```
NETSEC-ASA(config)#
NETSEC-ASA(config)#object network INSIDE-NET
NETSEC-ASA(config-network-object)#subnet 192.168.1.0 255.255.255.0
NETSEC-ASA(config-network-object)#nat (INSIDE,OUTSIDE) dynamic interface
NETSEC-ASA(config-network-object)#exit
NETSEC-ASA#show run
Saved
ASA Version 9.6(1)
!
hostname NETSEC-ASA
domain-name netsec.com
enable password 57n/mtd4HvB/bqUS encrypted
names
!
interface GigabitEthernet1/1
 nameif OUTSIDE
 security-level 0
 ip address 209.165.200.226 255.255.255.248
!
interface GigabitEthernet1/2
 nameif INSIDE
 security-level 100
 ip address 192.168.1.1 255.255.255.0
!
interface GigabitEthernet1/3
 no nameif
 no security-level
 no ip address
 shutdown
!
interface GigabitEthernet1/4
 no nameif
 no security-level
 no ip address
 shutdown
!
interface GigabitEthernet1/5
 no nameif
 no security-level
 no ip address
 shutdown
!
```

Fig 2.4 creating network object named INSIDE-NET

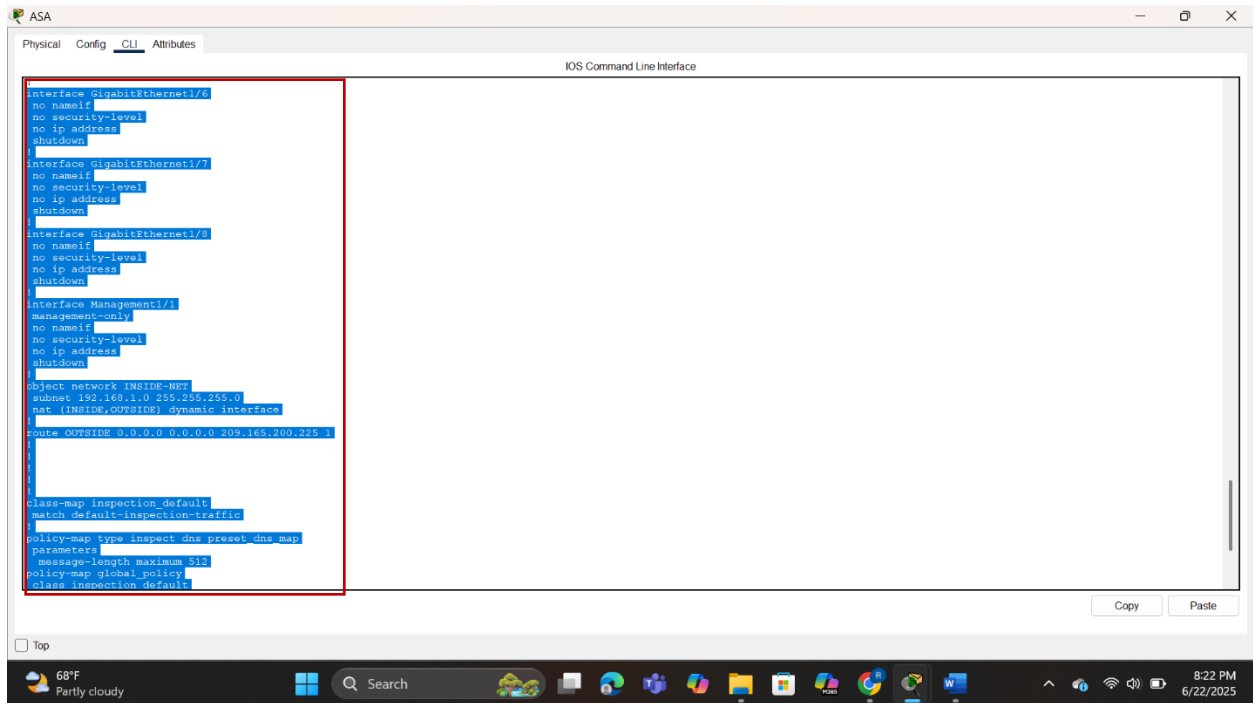


Fig 2.5 displaying NAT object configuration

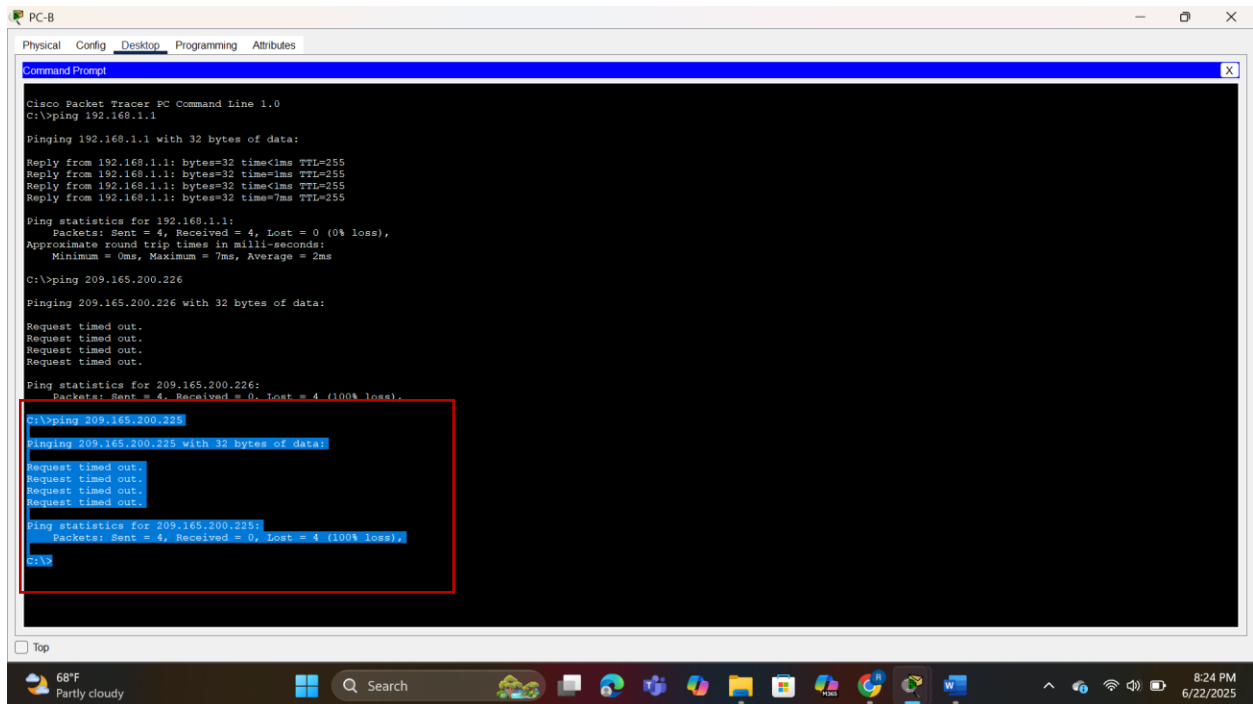


Fig 2.6 pinging R1 from pc B

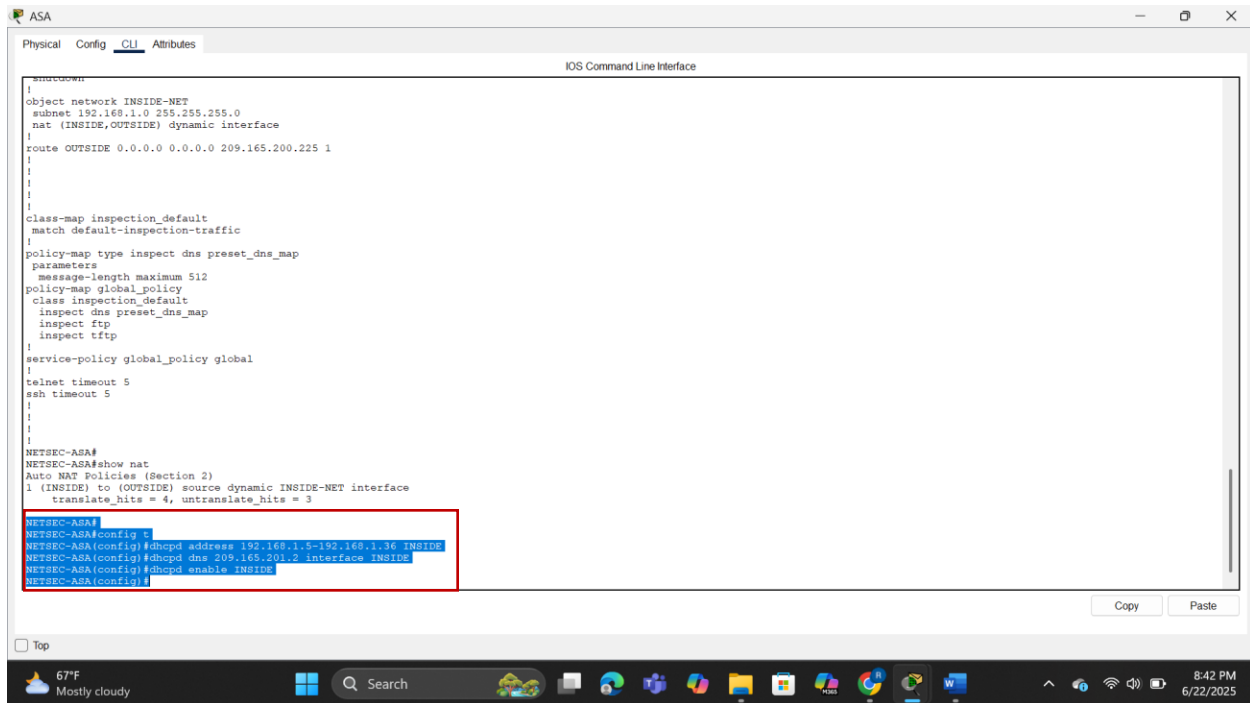


Fig 2.8 configuring ASA as DHCP server

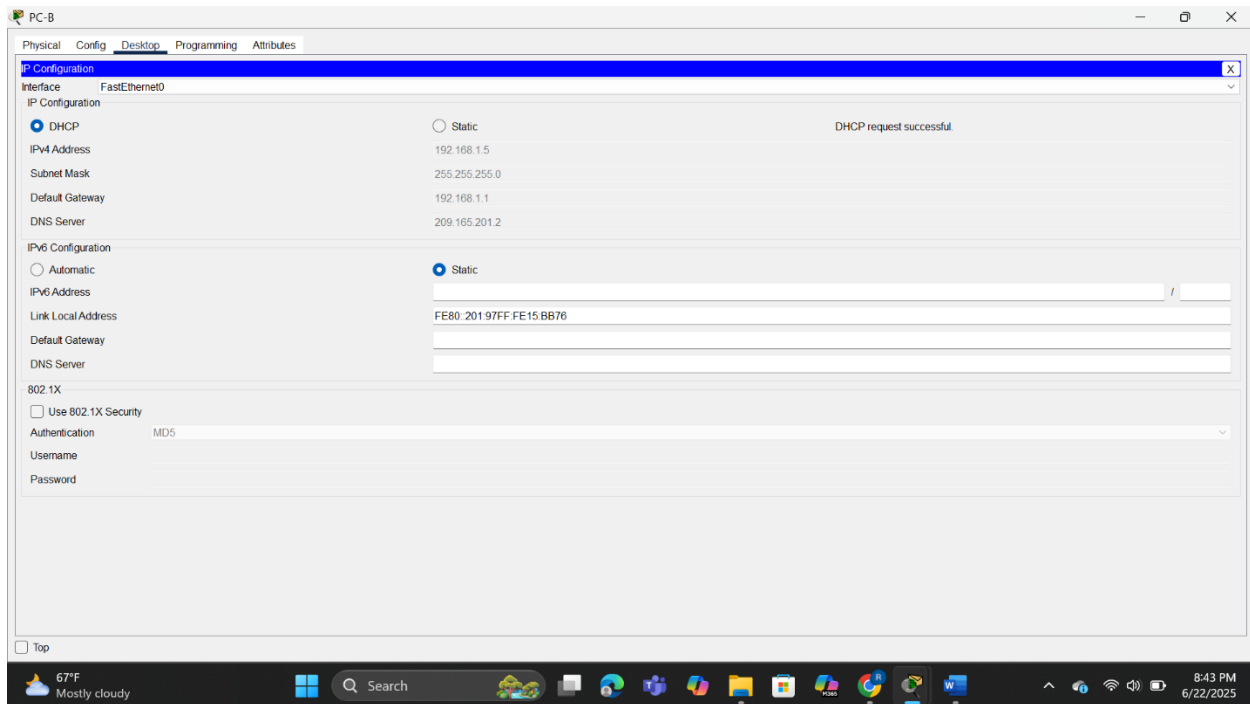
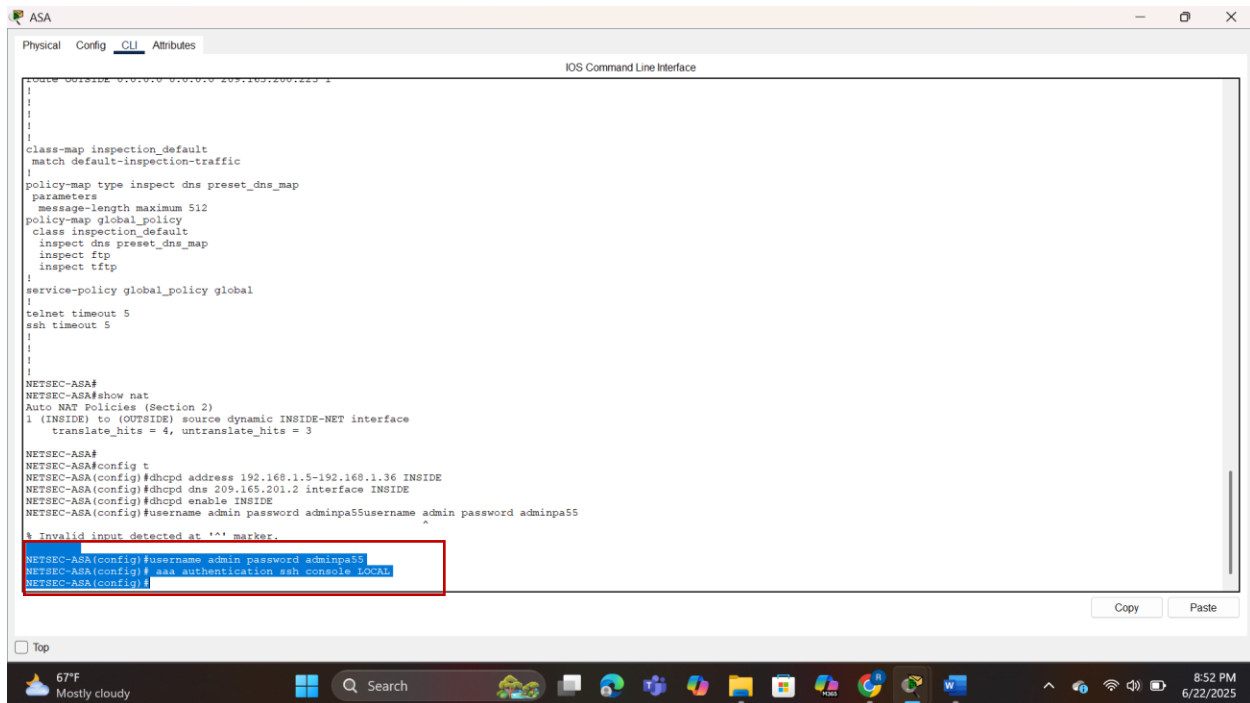


Fig 2.9 changing Pc B to receive ip dynamically(it receives an ip successfully)

Configuring AAA to use the local database for authentication

Next, I implemented AAA (Authentication, Authorization, and Accounting) to secure remote access. I created a local user account using username admin password adminpa55, defining a username "admin" with the password "adminpa55". To instruct the ASA to use this local database for authenticating SSH access, I entered the command `aaa authentication ssh console LOCAL`, ensuring any SSH session would prompt for credentials and verify them locally.



```
ASA
Physical Config CLI Attributes
IOS Command Line Interface

!
!
!
class-map inspection_default
  match default-inspection-traffic
!
!
policy-map type inspect dns preset_dns_map
  parameters
    message-length maximum 512
  policy-map global_policy
    class inspection_default
      inspect dns preset_dns_map
      inspect ftp
      inspect tftp
!
service-policy global_policy global
!
!
telnet timeout 5
ssh timeout 5
!
!
!
!
NETSEC-ASA#
NETSEC-ASA#show nat
Auto NAT Policies (Section 2)
1 (INSIDE) to (OUTSIDE) source dynamic INSIDE-NET interface
  translate_hits = 4, untranslate_hits = 3
!
NETSEC-ASA#
NETSEC-ASA#config t
NETSEC-ASA(config)#dhcpd address 192.168.1.5-192.168.1.36 INSIDE
NETSEC-ASA(config)#dhcpd dns 209.165.201.2 interface INSIDE
NETSEC-ASA(config)#dhcpd enable INSIDE
NETSEC-ASA(config)#username admin password adminpa55username admin password adminpa55
! Invalid input detected at '^' marker.
NETSEC-ASA(config)#username admin password adminpa55
NETSEC-ASA(config)#aaa authentication ssh console LOCAL
NETSEC-ASA(config)#
```

Fig 3.0 Configuring AAA to use the local database for authentication

Configuring remote access to the ASA

To enable secure remote access, I configured the ASA to accept SSH connections. I first attempted to generate an RSA key pair by issuing `crypto key generate rsa modulus 1024`. Since a key pair already existed, I declined to overwrite it by responding "no" when prompted. With key generation confirmed, I allowed SSH access from two sources. First, I permitted SSH from the entire INSIDE network using `ssh 192.168.1.0 255.255.255.0 INSIDE`, and second, I allowed SSH from the remote administrator located at 172.16.3.3 on the OUTSIDE using `ssh 172.16.3.3 255.255.255.255 OUTSIDE`. To ensure users wouldn't be logged out too quickly, I increased the SSH idle timeout to 10 minutes with the command `ssh timeout 10`.

To test SSH access, I launched a command prompt on PC-C (the remote administrator system) and connected to the ASA using `ssh -l admin 209.165.200.226`. The connection prompted for the configured credentials and successfully authenticated. Similarly, I tested internal SSH access from PC-B by entering `ssh -l admin 192.168.1.1`, and the connection was established as

expected. This confirmed that both inside and outside SSH access to the ASA was correctly configured, secured, and functional.

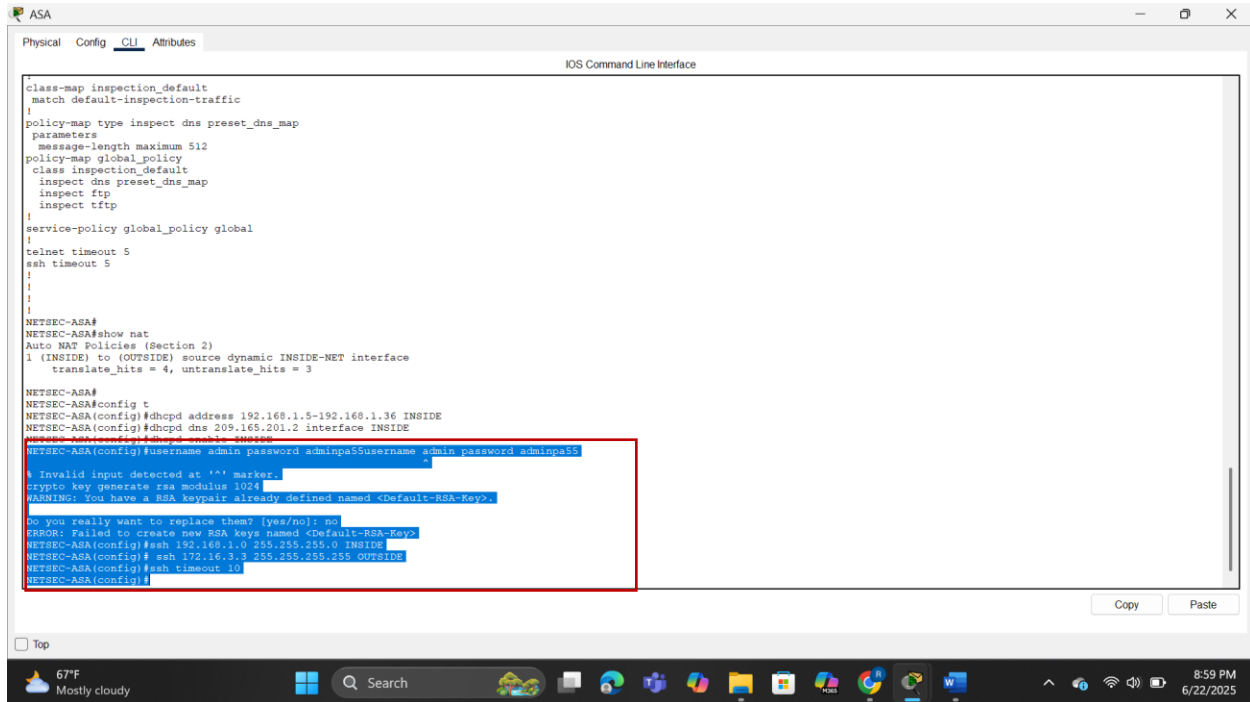


Fig 3.1 configuring remote access to the ASA

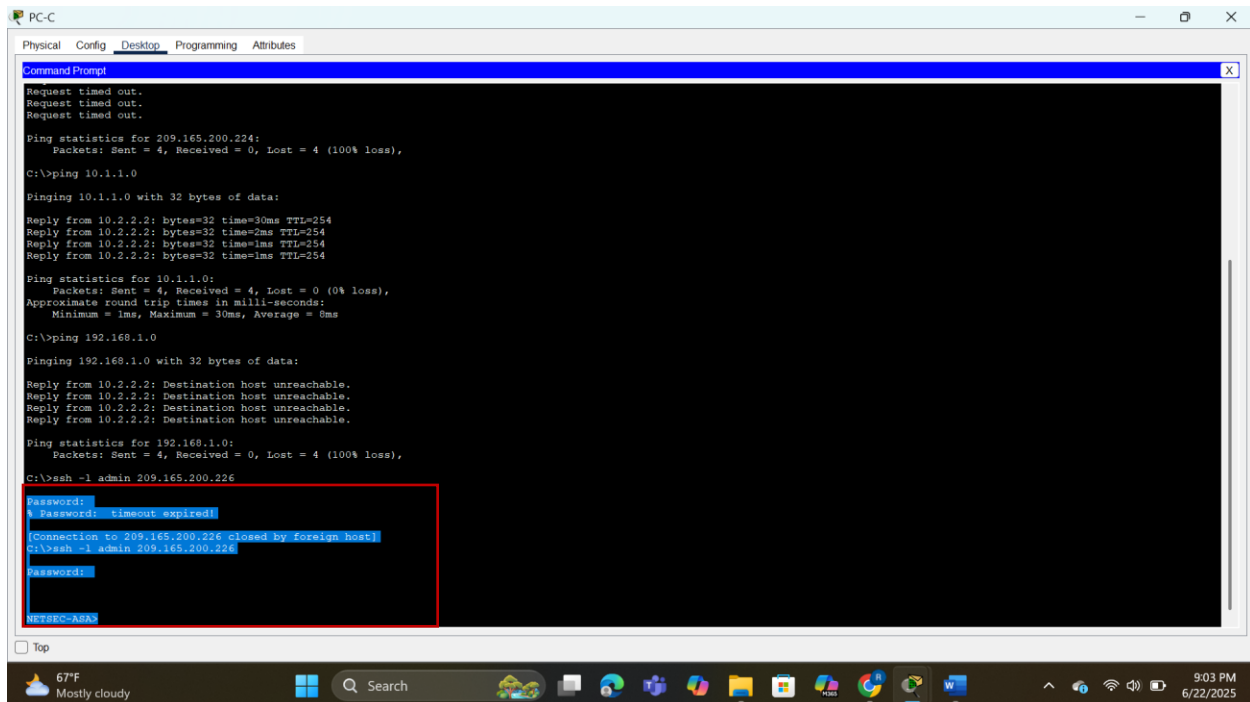


Fig 3.2 Establishing an SSH session from PC-C to the ASA

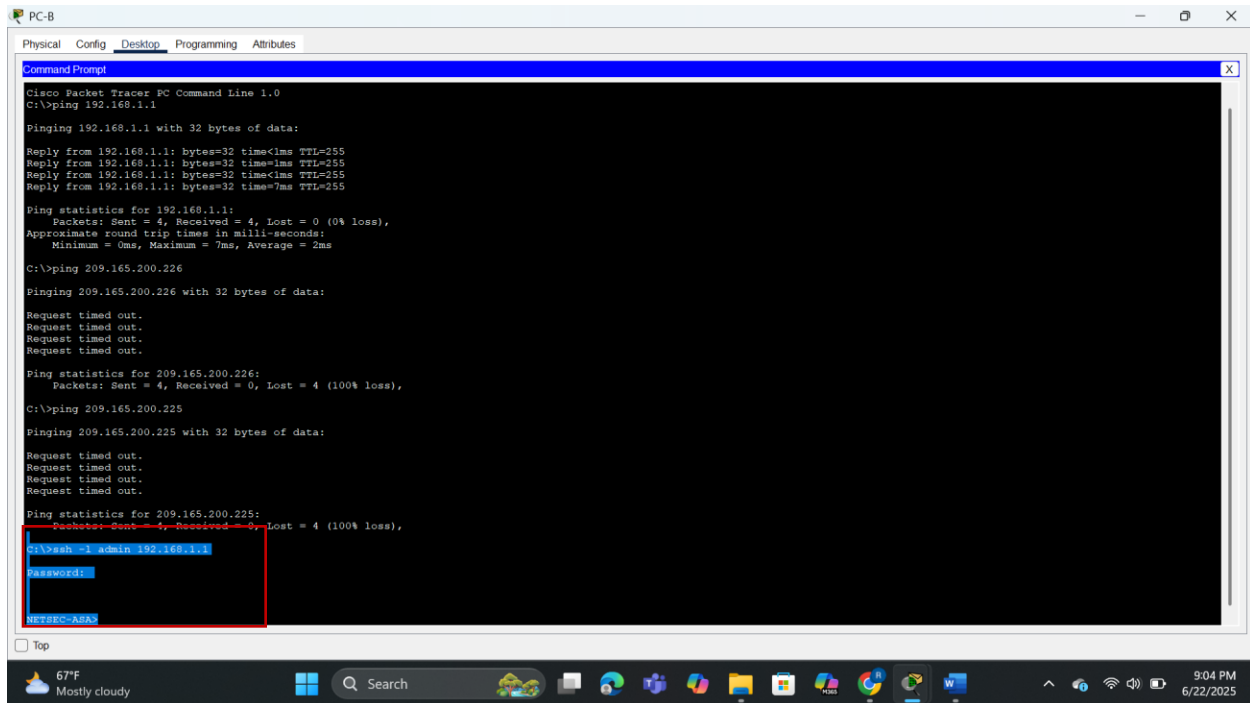


Fig 3.3 Establishing an SSH session from PC-B

Part 5

Configuring DMZ, Static NAT, and ACLs

Configuring the DMZ interface VLAN 3 on the ASA

To set up the DMZ, I first configured interface G1/3 on the ASA. I entered interface configuration mode with interface g1/3, then assigned the IP address 192.168.2.1 with subnet mask 255.255.255.0 using ip address 192.168.2.1 255.255.255.0. I named the interface “DMZ” with nameif DMZ, and by default, ASA assigned it a security level of 0. Since the DMZ should have more trust than the external network but less than the internal, I changed its security level to 70 using security-level 70. To activate the interface, I issued no shutdown. After configuring the DMZ interface, I verified the setup by running show interface ip brief and show ip address, which confirmed that the IP address, interface name, and status were correctly applied and that the interface was up.

```
message-length maximum 512
policy-map global_policy
class inspection_default
inspect dns preset_dns_map
inspect ftp
inspect tftp
!
service-policy global_policy global
!
telnet timeout 5
ssh timeout 5
!
!
!
!
NETSEC-ASA#
NETSEC-ASA#show nat
Auto NAT Policies (Section 2)
1 (INSIDE) to (OUTSIDE) source dynamic INSIDE-NET interface
  translate_hits = 4, untranslate_hits = 3
NETSEC-ASA#
NETSEC-ASA#config t
NETSEC-ASA(config)#dhcpd address 192.168.1.5-192.168.1.36 INSIDE
NETSEC-ASA(config)#dhcpd dns 209.165.201.2 interface INSIDE
NETSEC-ASA(config)#dhcpd enable INSIDE
NETSEC-ASA(config)#username admin password adminpa55username admin password adminpa55
% Invalid input detected at '^' marker.
crypto key generate rsa modulus 1024
WARNING: You have a RSA keypair already defined named <Default-RSA-Key>.
Do you really want to replace them? [yes/no]: no
ERROR: Failed to create new RSA keys named <Default-RSA-Key>
NETSEC-ASA(config)#ssh 192.168.1.0 255.255.255.0 INSIDE
NETSEC-ASA(config)# ssh 172.16.3.3 255.255.255.255 OUTSIDE
NETSEC-ASA(config)#ssh timeout 10
NETSEC-ASA(config)#
NETSEC-ASA(config)#interface g1/3
NETSEC-ASA(config-if)#ip address 192.168.2.1 255.255.255.0
NETSEC-ASA(config-if)#nameif DMZ
INFO: Security level for "DMZ" set to 0 by default.
NETSEC-ASA(config-if)#security-level 70
NETSEC-ASA(config-if)#no shutdown
```

Fig 3.4 Configuring the DMZ interface VLAN 3 on the ASA

```
NETSEC-ASA(config-if)#no shutdown
NETSEC-ASA(config-if)# show interface ip brief
Interface      IP-Address      OK? Method Status      Protocol
Virtual0       127.0.0.1        YES unset    up          up
GigabitEthernet1/1  209.165.200.226 YES manual  up          up
GigabitEthernet1/2  192.168.1.1      YES manual  up          up
GigabitEthernet1/3  192.168.2.1      YES manual  up          up
GigabitEthernet1/4  unassigned       YES unset  administratively down down
GigabitEthernet1/5  unassigned       YES unset  administratively down down
GigabitEthernet1/6  unassigned       YES unset  administratively down down
GigabitEthernet1/7  unassigned       YES unset  administratively down down
GigabitEthernet1/8  unassigned       YES unset  administratively down down
Management1/1     unassigned       YES unset  administratively down down
Internal-Controll/1 127.0.0.1.1      YES unset  up          up
Internal-Data1/1   unassigned       YES unset  up          up
Internal-Data1/2   unassigned       YES unset  up          up
Internal-Data1/3   unassigned       YES unset  up          up
NETSEC-ASA(config-if)# show ip address
System IP Addresses:
Interface      Name      IP address      Subnet mask      Method
GigabitEthernet1/1  OUTSIDE   209.165.200.226 255.255.255.248 manual
GigabitEthernet1/2  INSIDE    192.168.1.1     255.255.255.0    manual
GigabitEthernet1/3  DMZ       192.168.2.1     255.255.255.0    manual
GigabitEthernet1/4  unassigned unassigned      unassigned        unset
GigabitEthernet1/5  unassigned unassigned      unassigned        unset
GigabitEthernet1/6  unassigned unassigned      unassigned        unset
GigabitEthernet1/7  unassigned unassigned      unassigned        unset
GigabitEthernet1/8  unassigned unassigned      unassigned        unset
Management1/1     unassigned unassigned      unassigned        unset
Current IP Addresses:
Interface      Name      IP address      Subnet mask      Method
GigabitEthernet1/1  OUTSIDE   209.165.200.226 255.255.255.248 manual
GigabitEthernet1/2  INSIDE    192.168.1.1     255.255.255.0    manual
GigabitEthernet1/3  DMZ       192.168.2.1     255.255.255.0    manual
GigabitEthernet1/4  unassigned unassigned      unassigned        unset
GigabitEthernet1/5  unassigned unassigned      unassigned        unset
GigabitEthernet1/6  unassigned unassigned      unassigned        unset
GigabitEthernet1/7  unassigned unassigned      unassigned        unset
GigabitEthernet1/8  unassigned unassigned      unassigned        unset
Management1/1     unassigned unassigned      unassigned        unset
NETSEC-ASA(config-if)#
```

Fig 3.5 verifying configurations

Configuring static NAT to the DMZ server using a network object

Next, I configured static NAT for the DMZ server. I began by creating a network object named DMZ-SERVER using object network DMZ-SERVER. Within object configuration mode, I assigned it the internal IP address 192.168.2.3 using host 192.168.2.3. Then, I mapped this private address to the public IP 209.165.200.227 using static NAT with the command nat (DMZ,OUTSIDE) static 209.165.200.227, and then exited object configuration mode. This allowed external users to access the internal DMZ server via a public address.

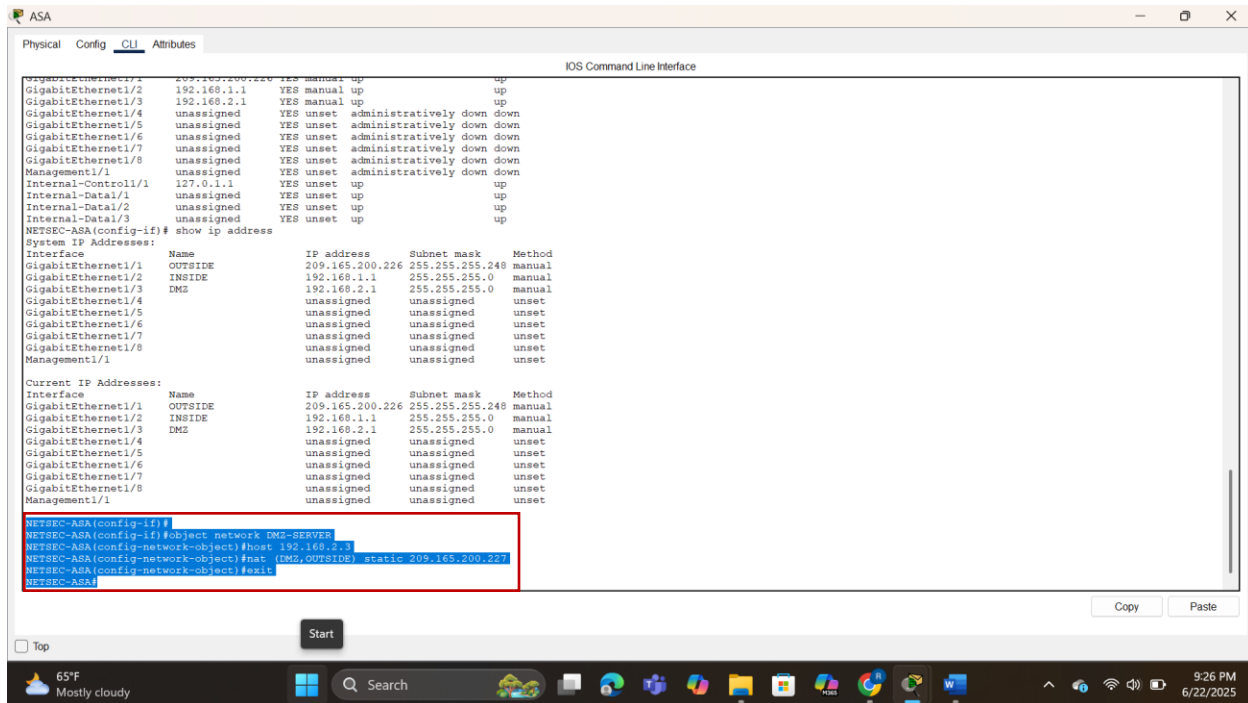


Fig 3.6 Configuring static NAT to the DMZ server using a network object.

Configuring an ACL to allow access to the DMZ server from the Internet

To control access to the DMZ server from the internet, I created an access control list. I named the ACL OUTSIDE-DMZ and configured it to permit two types of traffic: ICMP for pings using access-list OUTSIDE-DMZ permit icmp any host 192.168.2.3, and HTTP (port 80) using access-list OUTSIDE-DMZ permit tcp any host 192.168.2.3 eq 80. I applied this access list to the OUTSIDE interface in the inbound direction using access-group OUTSIDE-DMZ in interface OUTSIDE. It's important to note that in ASA ACLs, even though users connect to the public NAT address, the ACL references the internal IP (192.168.2.3) because NAT translation happens before ACLs are evaluated.

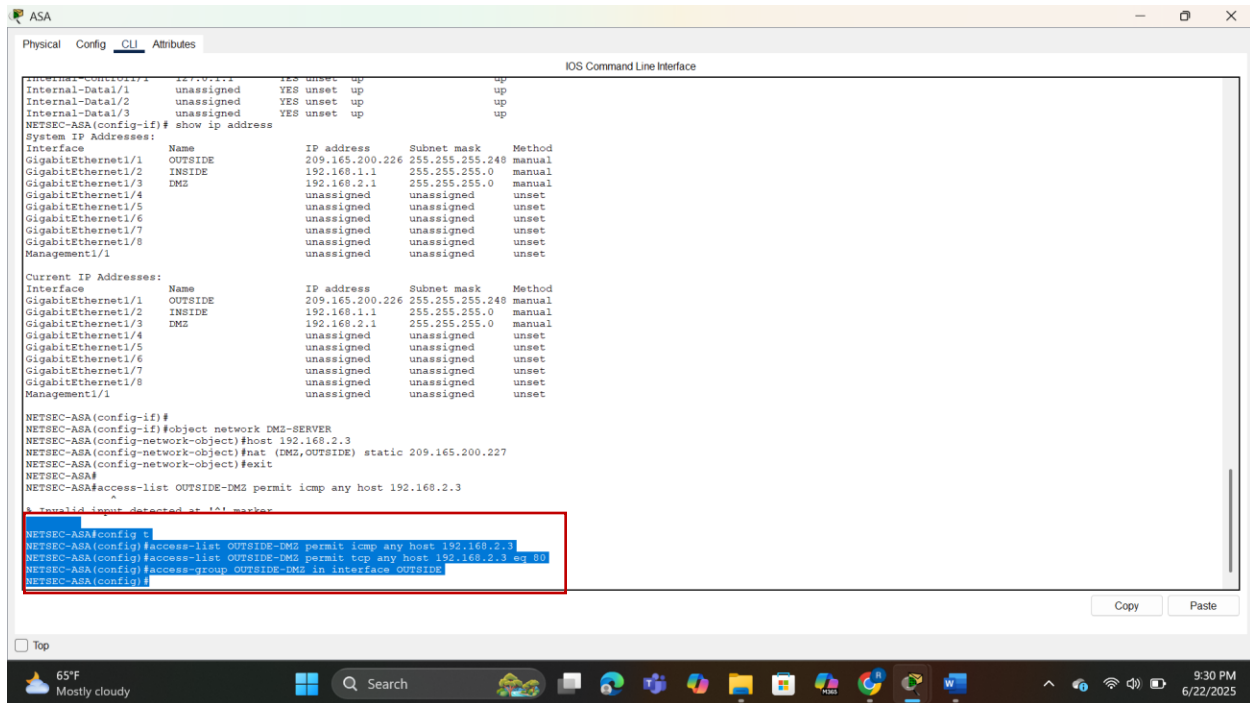


Fig 3.7 Configuring an ACL to allow access to the DMZ server from the Internet

Testing access to the DMZ server

Finally, to confirm the DMZ setup and NAT were functioning correctly, I went to PC-C (on the external 172.16.3.0/24 network) and opened a web browser. I entered the address <http://209.165.200.227>, which redirected to the DMZ server via NAT. The page loaded successfully, verifying that external HTTP access was correctly routed through the ASA to the DMZ server, completing the setup of a secure and functional demilitarized zone.

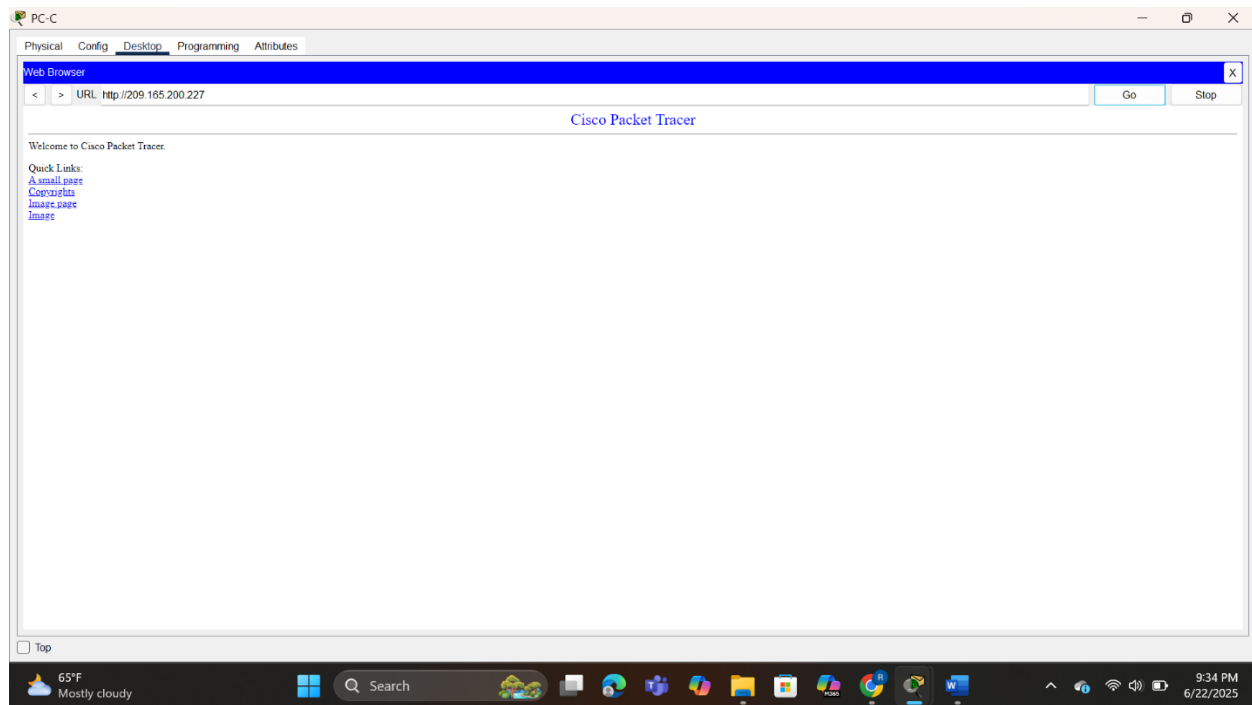


Fig 3.8 Testing access to the DMZ server.

Conclusion

Through the hands-on configuration of the ASA firewall using CLI, I successfully implemented a secure, multi-zoned network environment. Each component—ranging from interface security levels, NAT policies, and access lists, to DHCP and remote management via SSH—was configured in alignment with best practices for layered defense. The creation of a DMZ and implementation of static NAT demonstrated how controlled public access to internal resources can be securely managed. The structured approach to configuring and testing connectivity and firewall rules provided a comprehensive understanding of ASA's role in network security. Ultimately, this lab strengthened my technical skills in firewall management and highlighted the importance of precision, verification, and security-focused design in managing real-world enterprise networks.